

Supplementary Material: A Bayesian Multiscale Integrated Abundance Model for Estimating Latent Opioid Misuse Prevalence from Spatially Misaligned Data

Staci A. Hepler, Brian N. White, Magdalena Cerdá,
William C. Miller, Lance A. Waller, and David M. Kline

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1 Supplemental Data

1.1 State-Level Data

Year	Estimate	SE
2007 - 2010	0.0550	0.0024
2011 - 2014	0.0520	0.0026
2015 - 2016	0.0470	0.0041
2016 - 2017	0.0510	0.0037
2017 - 2018	0.0410	0.0033
2018 - 2019	0.0430	0.0043
2021 - 2022	0.0340	0.0040

Table 1: NSDUH estimates and standard errors (SE) of past-year opioid misuse included in analysis.

1.2 County-Level Data

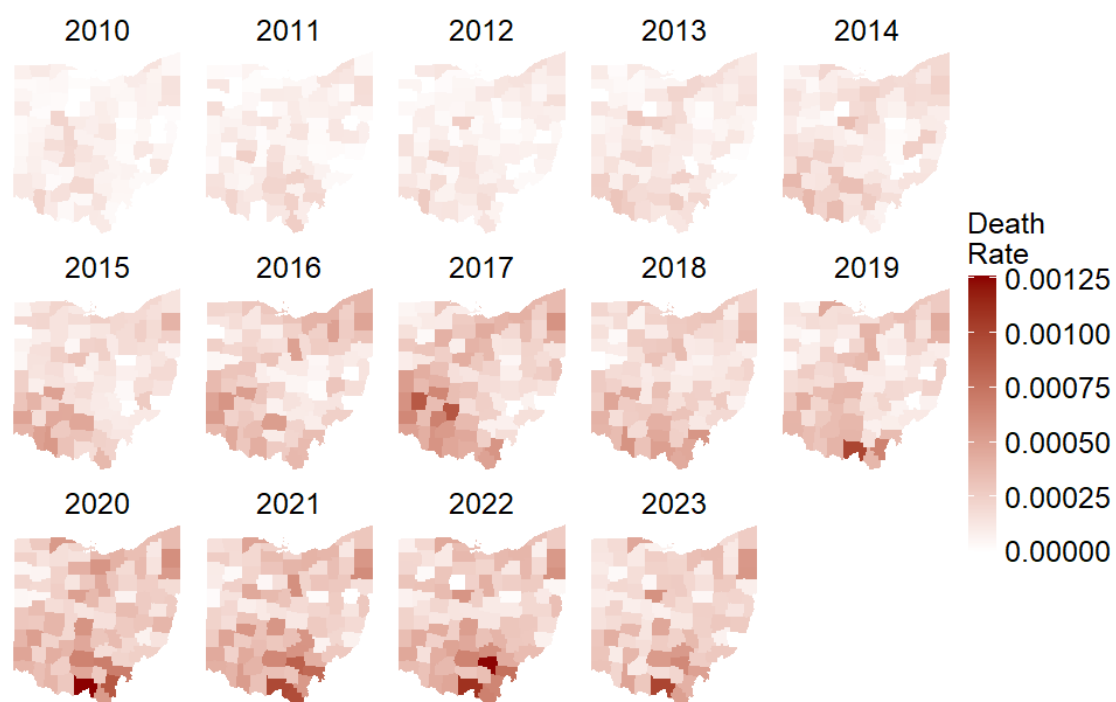


Figure 1: Observed county-level opioid overdose death rates per population for 2010 to 2023.

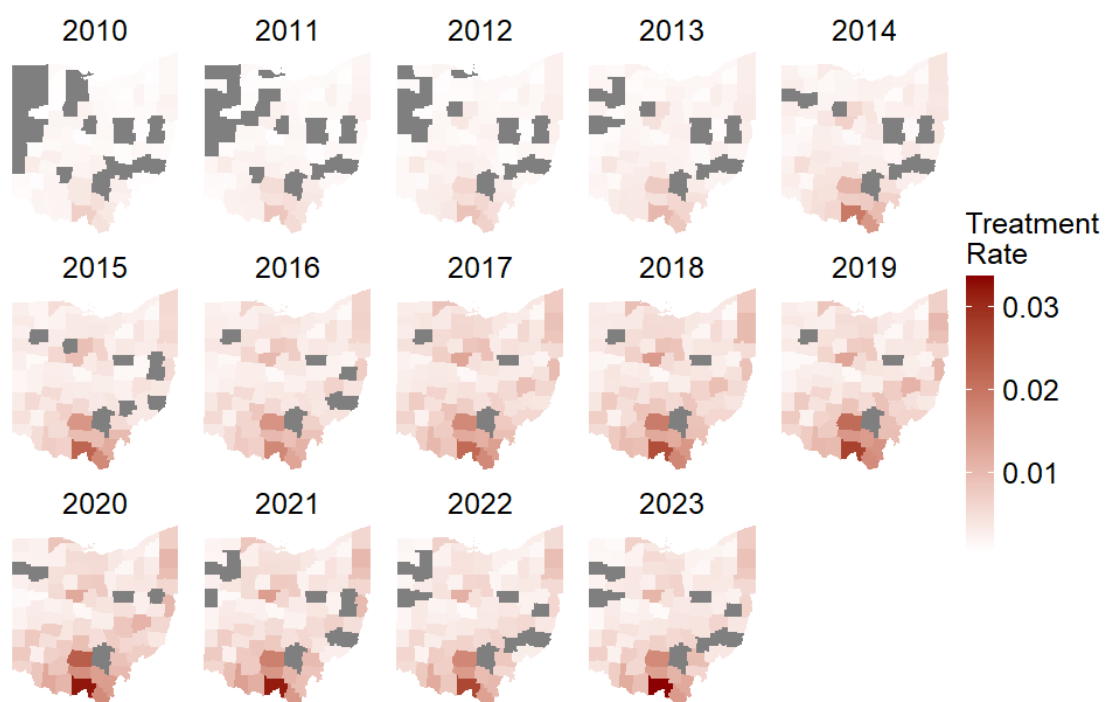


Figure 2: Observed county-level opioid treatment rates per population for 2010 to 2023. Note that counts less than 50 were suppressed, and the rates corresponding to counties with suppressed values are shown as gray on the map.

1.3 ZCTA-Level Data

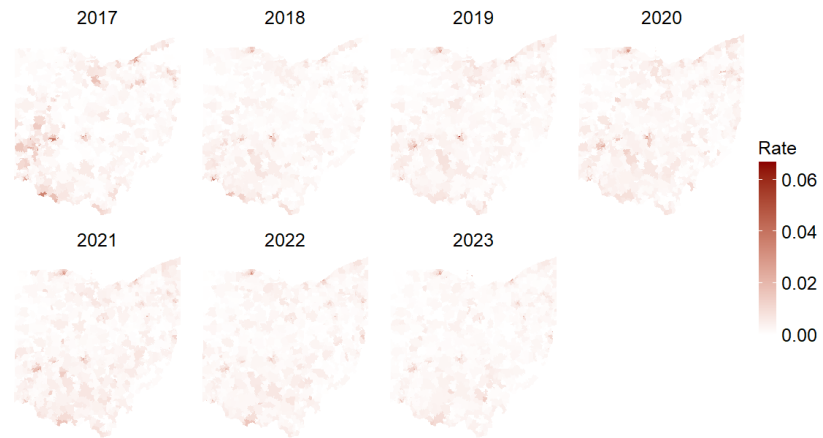


Figure 3: Observed ZCTA-level rates of EMS administered naloxone per population for 2017 to 2023.

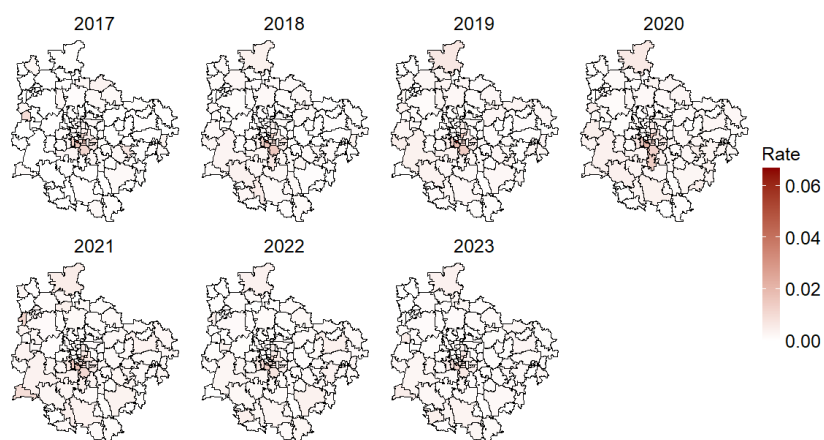


Figure 4: Observed ZCTA-level rates of EMS administered naloxone per population for 2017 to 2023, zoomed in to only include ZCTA that intersect with central Ohio counties.

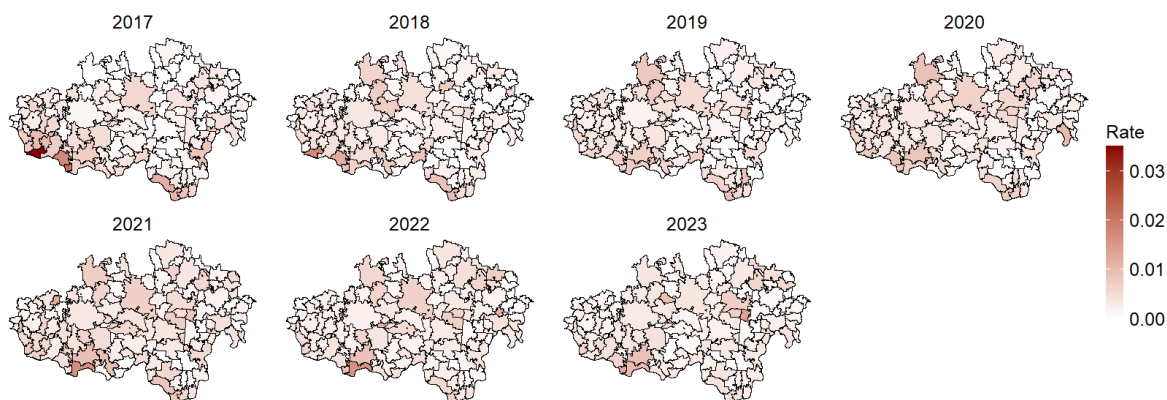


Figure 5: Observed ZCTA-level rates of EMS administered naloxone per population for 2017 to 2023, zoomed in to only include ZCTA that intersect with southern Ohio counties.

1.4 Atom-Level Data

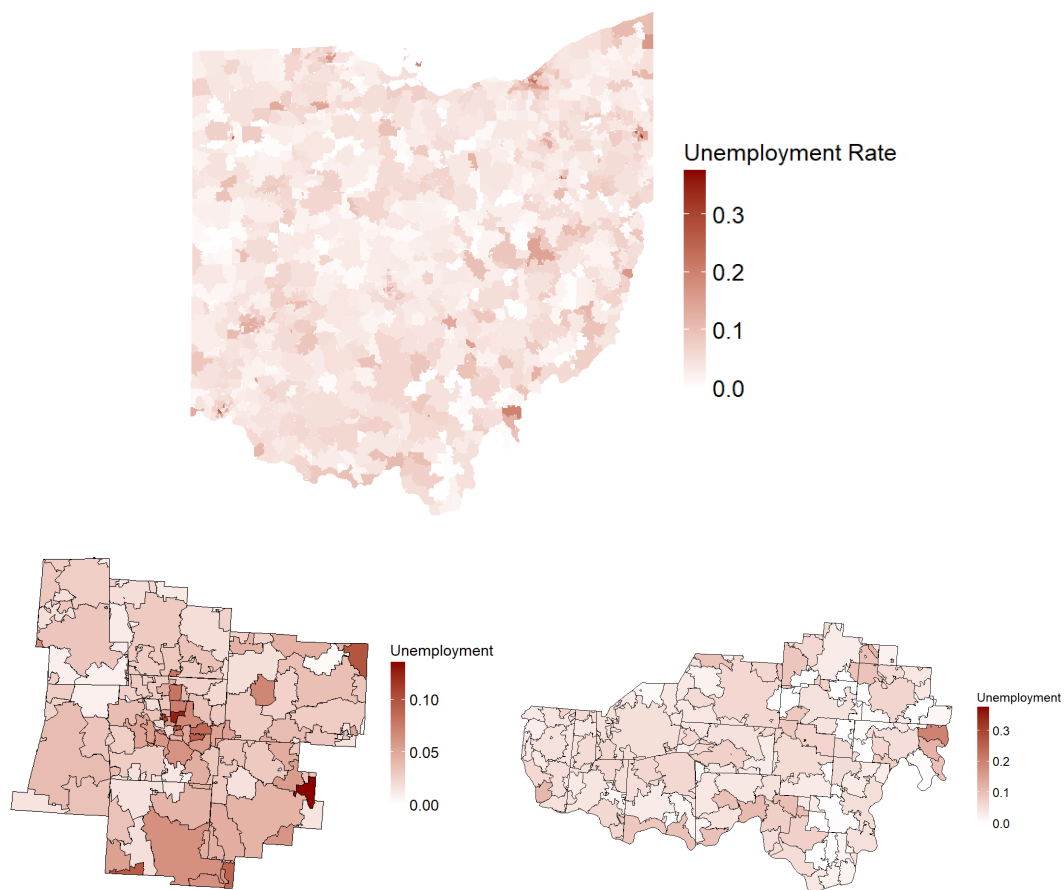


Figure 6: Proportion of the atom's population that is unemployed. The top map is for all atoms in Ohio, the bottom left map zooms into central Ohio, and the bottom right map shows atoms in southern Ohio.

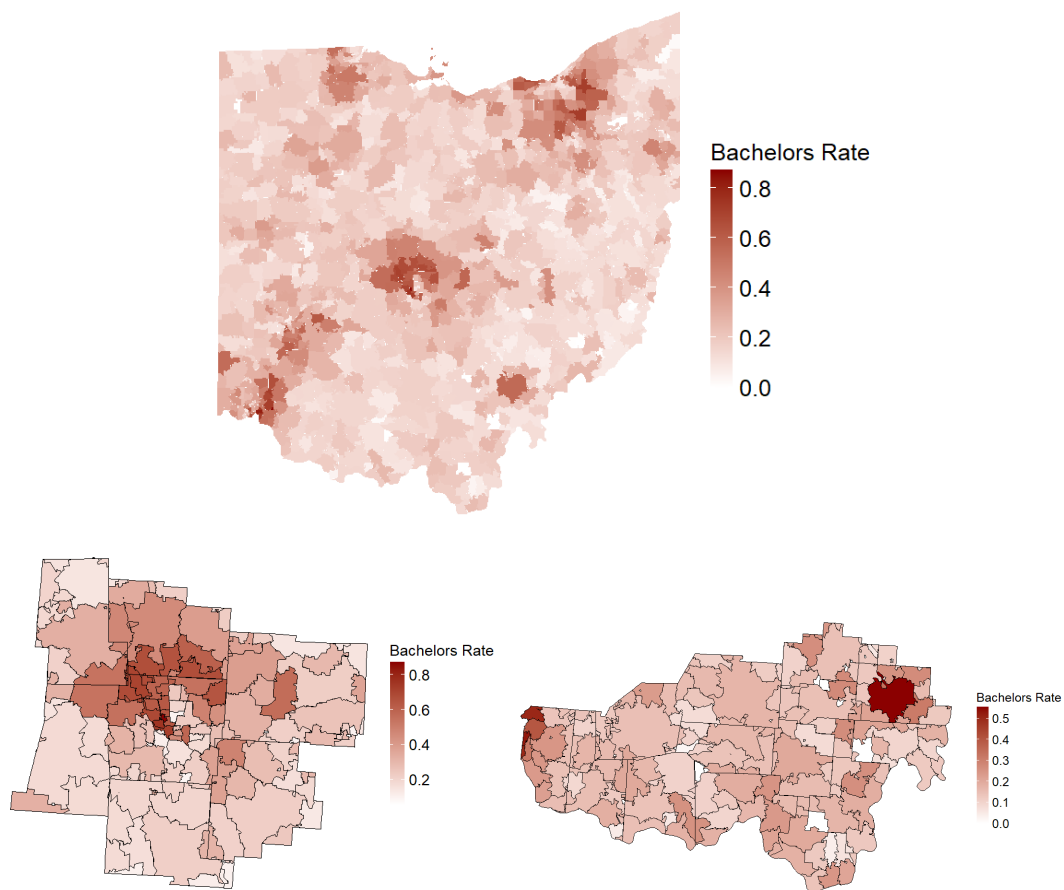


Figure 7: Proportion of the atom's population with at least a bachelor's degree. The top map is for all atoms in Ohio, the bottom left map zooms into central Ohio, and the bottom right map shows atoms in southern Ohio.

2 Supplemental Modeling Details

2.1 County-Level Bayesian Integrated Abundance Model

The Bayesian Integrated Abundance (BIA) model that is the foundation of the proposed MIA model is briefly reviewed in the main text, but the full model is described in detail below.

Let $Y_{it}^{(k)}$ denote the observed count for indicator $k = 1, \dots, K$ at county $i = 1, \dots, n^{(C)}$ during year $t = 1, \dots, T$. In our application, $k = 1$ corresponds to opioid overdose deaths and $k = 2$ corresponds to treatment admissions. Conditional on the county-level latent number of PWMO, N_{it} , we assume

$$Y_{it}^{(k)} | N_{it}, p_{it}^{(k)} \stackrel{ind}{\sim} \text{Binomial} \left(N_{it}, p_{it}^{(k)} \right), \quad (1)$$

where $p_{it}^{(k)}$ represents the probability that an individual who misuses opioids is counted by indicator k . We model the indicator probabilities using logistic regression

$$\text{logit} \left(p_{it}^{(k)} \right) = \mu_t^{(k)} + \mathbf{X}_{it}^{(k)} \boldsymbol{\beta}^{(k)} + f_{it}^{(k)} + \epsilon_{it}^{(k)},$$

where $\mu_t^{(k)}$ is a time-varying intercept capturing statewide temporal trends, $\mathbf{X}_{it}^{(k)}$ is a vector of centered covariates (excluding an intercept), $\boldsymbol{\beta}^{(k)}$ is a vector of regression coefficients, $f_{it}^{(k)}$ is a spatio-temporal random effect, and $\epsilon_{it}^{(k)}$ accounts for additional unexplained heterogeneity. Note that in our application, treatment admission counts, $Y_{it}^{(2)}$, that are less than 50 are censored. We account for this in the model using an interval censored likelihood as in Famoye & Wang (2004). More specifically, let

$$c_{it} = \begin{cases} 0 & \text{treatment count observed} \\ 1 & \text{treatment count censored.} \end{cases} \quad (2)$$

For $c_{it} = 1$, we know the actual treatment count in county i during year t must be between 0 and 49. The likelihood is then

$$\begin{aligned} L(\mathbf{p}^{(1)}, \mathbf{p}^{(2)}, \mathbf{N} | \mathbf{Y}^{(1)}, \mathbf{Y}^{(2)}) &= \left[\prod_{t=1}^T \prod_{i=1}^{n^{(C)}} f \left(Y_{it}^{(1)} | p_{it}^{(1)}, N_{it} \right) \right] \\ &\times \left[\prod_{t=1}^T \prod_{i=1}^{n^{(C)}} \left[f \left(Y_{it}^{(2)} | p_{it}^{(2)}, N_{it} \right) \right]^{1-c_{it}} \left[F \left(49 | p_{it}^{(2)}, N_{it} \right) - F \left(0 | p_{it}^{(2)}, N_{it} \right) \right]^{c_{it}} \right] \end{aligned}$$

where $f(\cdot|p_{it}^{(1)}, N_{it})$ is the probability mass function of the binomial distribution with population size N_{it} and rate $p_{it}^{(1)}$, and $F(\cdot|p_{it}^{(2)}, N_{it})$ is the cumulative distribution function and $f(\cdot|p_{it}^{(2)}, N_{it})$ the probability mass function of the binomial distribution with population size N_{it} and rate $p_{it}^{(2)}$.

To capture spatial and temporal dependence, we assume the random effects follow an intrinsic conditional autoregressive (ICAR) model across counties, coupled with first-order temporal autoregression. Specifically, for $t = 1$

$$f_{it}^{(k)}|\mathbf{f}_{-i,t}^{(k)}, \tau_k^2 \sim N\left(\frac{1}{w_{i+}} \sum_j w_{ij} f_{jt}^{(k)}, \frac{\tau_k^2}{w_{i+}}\right), \quad (3)$$

and for $t = 2, \dots, T$

$$f_{it}^{(k)}|\mathbf{f}_{-i,t}^{(k)}, f_{i,t-1}^{(k)}, \tau_k^2, \phi_k \sim N\left(\phi_k f_{i,t-1}^{(k)} + \frac{1}{w_{i+}} \sum_j w_{ij} (f_{jt}^{(k)} - \phi_k f_{j,t-1}^{(k)}), \frac{\tau_k^2}{w_{i+}}\right), \quad (4)$$

where $\mathbf{f}_{-i,t}^{(k)} = \{f_{jt}^{(k)} : j \neq i\}$, w_{ij} is an indicator that counties i and j share a border, $w_{i+} = \sum_j w_{ij}$ is the total number of neighbors for county i , τ_k^2 is related to the conditional variance, and ϕ_k is a temporal autocorrelation parameter. To capture between-outcome residual correlation, we assume each $\boldsymbol{\epsilon}_{it} = (\epsilon_{it}^{(1)}, \dots, \epsilon_{it}^{(K)})$ independently has a K -dimensional multivariate normal distribution with mean zero and covariance matrix Σ_ϵ .

A challenge in this setting is that surveillance data alone do not identify the overall level of OMP (Stoner et al. 2019). To address this, we incorporate data from statewide National Survey on Drug Use and Health (NSDUH) estimates, which provide estimates of statewide OMP over multi-year periods, along with standard errors of the estimates. Let $S_{a:b}$ denote the estimate of the statewide OMP between years a and b . The estimates for Ohio used in our application are shown in Table 1. We apply a logit-transformation to the estimates and use a first-order Taylor expansion to approximate the standard error. Let $\hat{se}_{a:b}$ denote the resulting standard error of the logit-transformed estimated OMP. Let μ_t denote the true latent statewide OMP in year t . We assume $\text{logit}(\mu_t) = \beta_0^\mu + \beta_1^\mu t$, and

$$\text{logit}(S_{a:b})|\beta_0^\mu, \beta_1^\mu \sim N\left(\frac{1}{b-a+1} \sum_{t=a}^b \text{logit}(\mu_t), \hat{se}_{a:b}^2\right). \quad (5)$$

The target of inference in the county-level model is the latent number of PWMO, N_{it} , and the prevalence ratio, λ_{it} . Assume

$$N_{it} | \lambda_{it} \stackrel{ind}{\sim} \text{Binomial}(P_{it}, \mu_t \lambda_{it}),$$

where P_{it} is the known population of county i during year t and μ_t is the statewide OMP that also appeared in (5). The prevalence ratio is relative to the statewide OMP and modeled as

$$\log(\lambda_{it}) = \mathbf{W}_{it}^{(C)} \boldsymbol{\gamma} + u_{it} + v_{it},$$

where $\mathbf{W}_t^{(C)}$ is a design matrix for time t containing centered covariates without an intercept, $v_{it} \stackrel{iid}{\sim} N(0, \sigma_v^2)$, and u_{it} is a spatio-temporal random effect defined analogously to (3)-(4) with ϕ_u a temporal autocorrelation parameter and τ_u^2 the conditional variance parameter.

All intercepts and regression coefficients are independently assigned diffuse mean-zero normal distributions with standard deviations of 100, and temporal autocorrelation parameters are assumed to be uniform over $(0, 1)$. All variance parameters are assigned inverse gamma prior distributions with shape and scale of 0.5. The residual covariance matrix $\boldsymbol{\Sigma}_\epsilon$ has an inverse Wishart prior with identity scale matrix and K degrees of freedom.

2.2 ZCTA-Level Outcome Model Additional Details

Recall, $Z_{\ell,t}^{(o)}$ denotes the observed count for ZCTA-level indicator o at ZCTA ℓ during year t , and we assume $Z_{\ell,t}^{(o)} = \sum_j Z_t^{(o)}(B_{\ell j})$, where $Z_t^{(o)}(B_{\ell j})$ denotes the *latent* count of indicator o for atom j within ZCTA ℓ . For each atom, assume

$$Z_t^{(o)}(B_{\ell j}) \mid N_t(B_{\ell j}), q_t^{(o)}(B_{\ell j}) \sim \text{Poisson} \left(N_t(B_{\ell j}) q_t^{(o)}(B_{\ell j}) \right),$$

where $N_t(B_{\ell j})$ is the latent atom-level count of PWMO in year t , and $q_t^{(o)}(B_{\ell j})$ is the rate at which PWMO are counted by indicator o in that atom during that year. We model $q_t^{(o)}(B_{\ell j})$ with a log-linear model of the form

$$\log(q_t^{(o)}(B_{\ell j})) = \mathbf{X}_t^{(o)}(B_{\ell j}) \boldsymbol{\beta}^{(o)} + f_t^{(o)}(B_{\ell j}), \quad (6)$$

where $\mathbf{X}_t^{(o)}(B_{\ell j})$ is a vector of atom-level covariates with regression coefficients $\boldsymbol{\beta}^{(o)}$ and $f_t^{(o)}(B_{\ell j})$ is a spatio-temporal random effect.

If $n^{(a)}$ is large, as in our application, an atom-level spatio-temporal random effect makes the model computationally expensive. We assume $f_t^{(o)}(B_{\ell j}) = \mathbf{B}(B_{\ell j})' \boldsymbol{\omega}_t^{(o)}$, where $\mathbf{B}(B_{\ell j})$ is a q -dimensional vector of Moran's I basis functions at atom $B_{\ell j}$ (Hughes & Haran 2013). More specifically, for our application the basis functions are the q eigenvectors of the Moran operator,

$$\left(\mathbf{I} - \mathbf{X}_1^{(o)} \left(\mathbf{X}_1^{(o)'} \mathbf{X}_1^{(o)} \right)^{-1} \mathbf{X}_1^{(o)'} \right) \mathbf{A} \left(\mathbf{I} - \mathbf{X}_1^{(o)} \left(\mathbf{X}_1^{(o)'} \mathbf{X}_1^{(o)} \right)^{-1} \mathbf{X}_1^{(o)'} \right),$$

corresponding to the q largest, positive eigenvalues, where $\mathbf{X}_1^{(o)}$ is the design matrix used in (6) in year $t = 1$ and $\mathbf{A}$ is the atom-level adjacency matrix. Note that the atom-level covariates used in our application are static, so $\mathbf{X}_t = \mathbf{X}_1$ for all t , but that does not have to hold generally.

The vector $\boldsymbol{\omega}_t^{(o)}$ is a q -dimensional vector of coefficients for the basis functions for indicator o at time t . We account for temporal dependence by assuming an autoregressive model of order one for each of the $i = 1, \dots, q$ coefficients. That is,

$$\begin{aligned} \omega_{1,i}^{(o)} &\sim N(0, \tau_{o,i}^2) \quad \text{for } t = 1 \\ \omega_{t,i}^{(o)} &\sim N(\phi_i^{(o)} \omega_{t-1,i}^{(o)}, \tau_{o,i}^2) \quad \text{for } t \geq 1, \end{aligned}$$

where $\tau_{o,i}^2$ are variance parameters and $\phi_i^{(o)}$ are autocorrelation parameters for each basis function $i = 1, \dots, q$ and each ZCTA-level outcome o .

3 Supplemental Simulation Data and Results

3.1 Simulated Latent Atom-Level Counts of Misuse

In this section, we show maps of the simulated latent atom-level prevalence of misuse, $N_t(A_{it})/P_t(A_{ij})$, for the 8th and 23rd simulated data sets. The atom-level counts were simulated assuming the FNCH distribution described in the main manuscript, and as such, the atoms within a county are forced to sum to the county-level value N_{it} , which is assumed to be the same for all 100 simulate data sets. Figure 8 shows maps for the entire state of Ohio. To more easily visualize the spatial heterogeneity, we show the same maps but zoomed in to only include central Ohio counties (Figure 9) and southern Ohio counties (Figure 10). Central Ohio counties are defined to include Franklin, Delaware, Licking, Fairfield, Pickaway, Madison, and Union counties. Southern Ohio counties include Adams,

Athens, Brown, Clermont, Gallia, Highland, Hocking, Jackson, Lawrence, Meigs, Pike, Ross, Scioto, and Vinton counties.

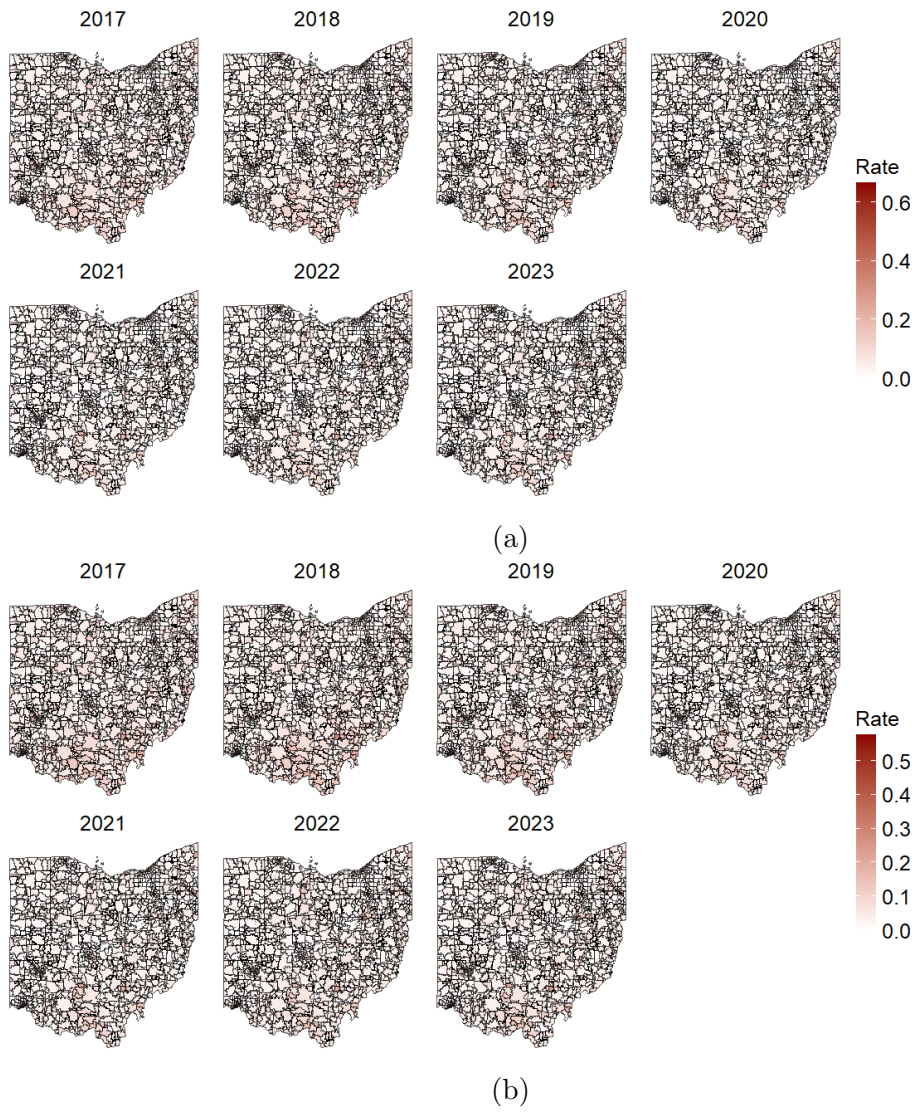
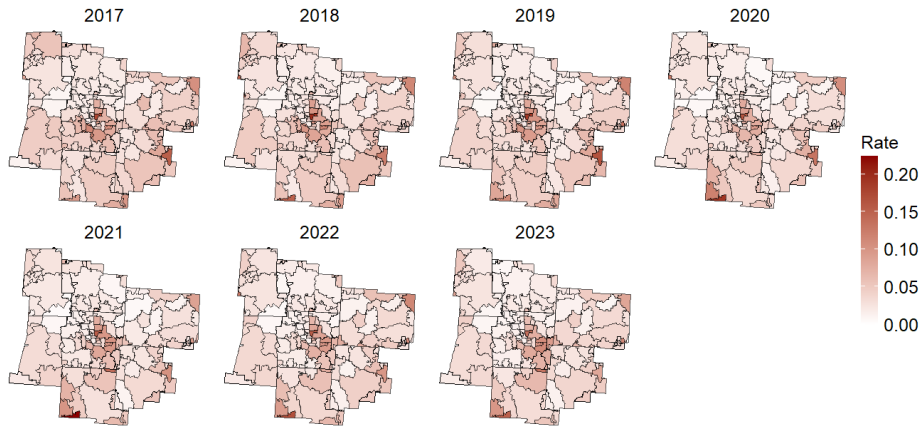
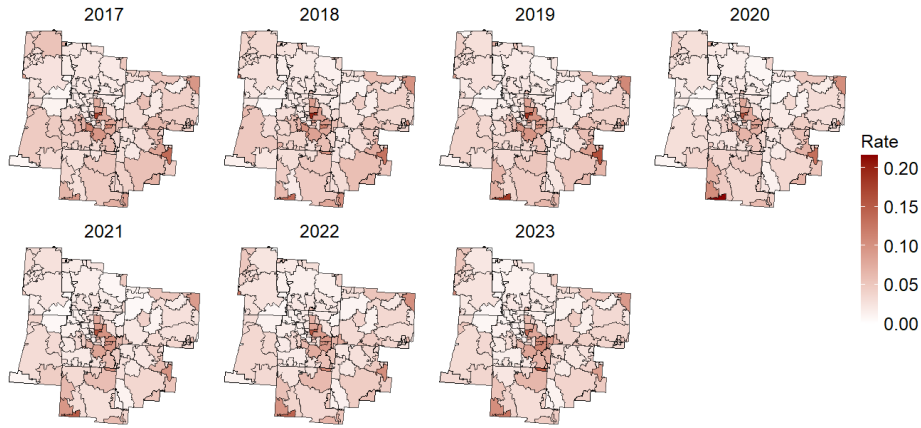


Figure 8: Simulated atom-level prevalence of misuse for the 8th (Figure a) and 23rd (Figure b) simulated data sets.

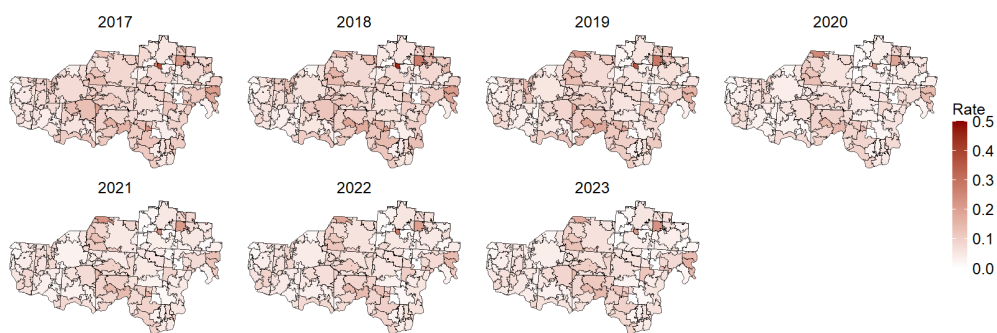


(a)

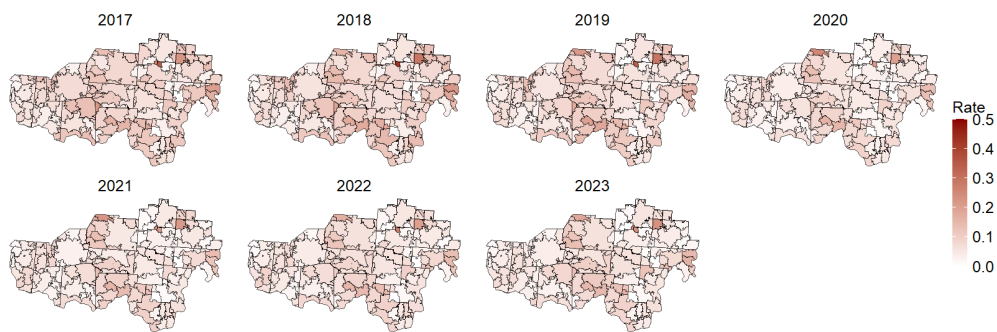


(b)

Figure 9: Simulated atom-level prevalence of misuse, zoomed into the atoms contained within central Ohio counties, for the 8th (Figure a) and 23rd (Figure b) simulated data sets.



(a)



(b)

Figure 10: Simulated atom-level prevalence of misuse, zoomed into the atoms contained within southern Ohio counties, for the 8th (Figure a) and 23rd (Figure b) simulated data sets.

3.2 Simulated ZCTA-level Counts of Naloxone Administrations

Figures 11 - 13 show maps of the ZCTA-level rates, $Z_{\ell t}/P_{\ell t}$. The ZCTA-level counts serve as the observed “data” in the simulation study. Maps are shown for the 8th and 23rd simulated data set. Figure 11 shows the map for the entire state of Ohio; whereas Figures 12 and 13 zoom in to show just the ZCTA that intersect with central Ohio and southern Ohio counties, respectively.

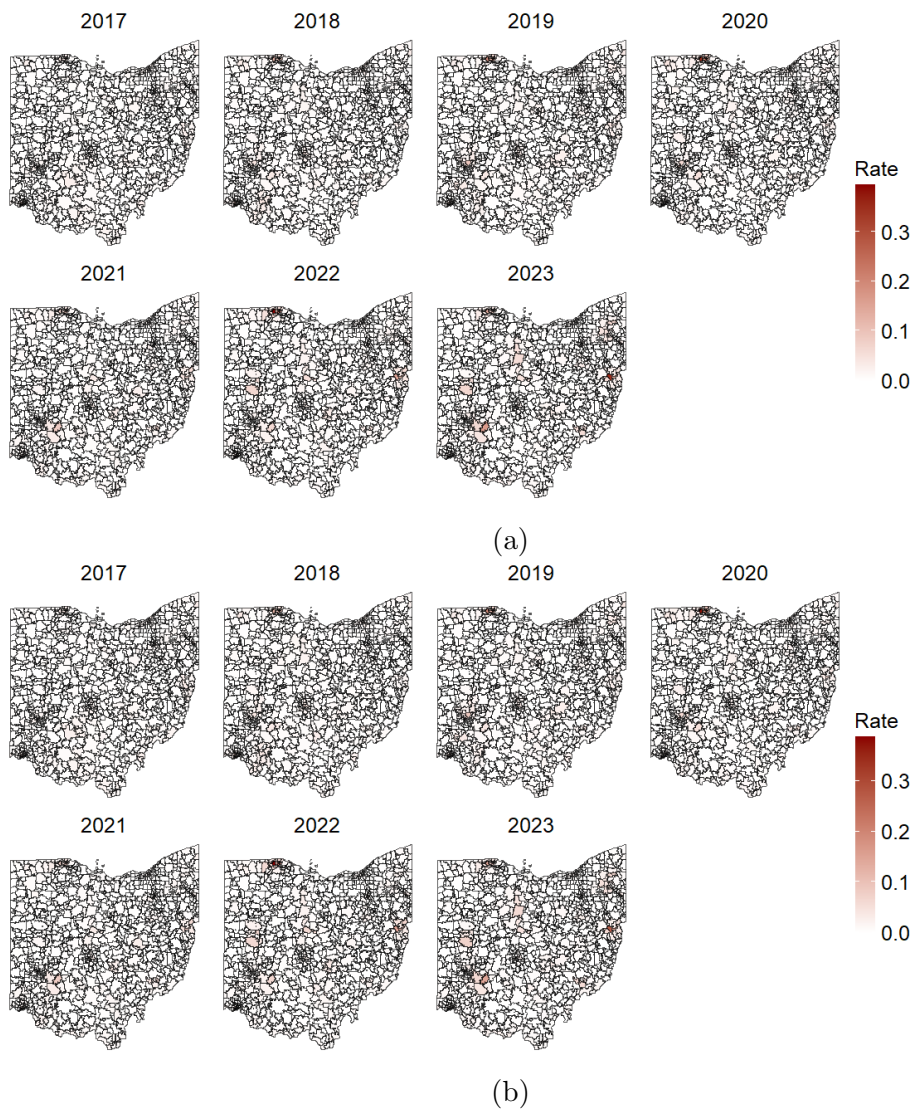
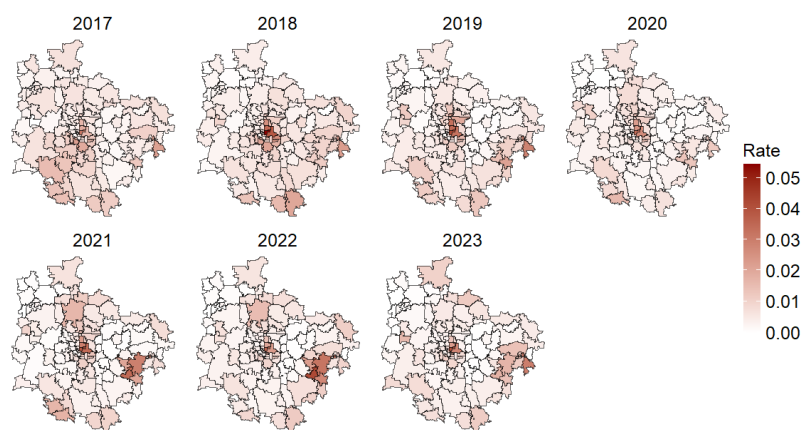
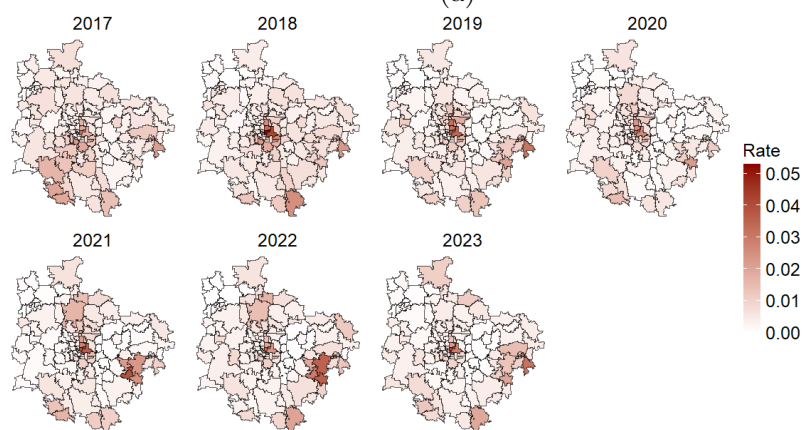


Figure 11: Simulated ZCTA-level rates of naloxone administrations for the 8th (Figure a) and 23rd (Figure b) simulated data sets.

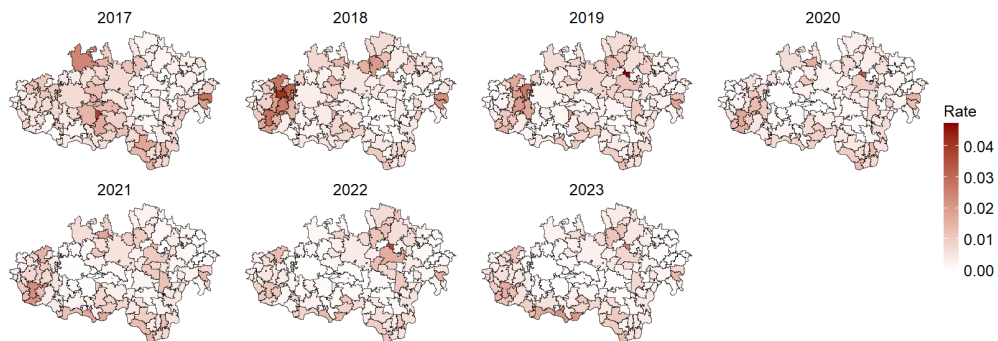


(a)

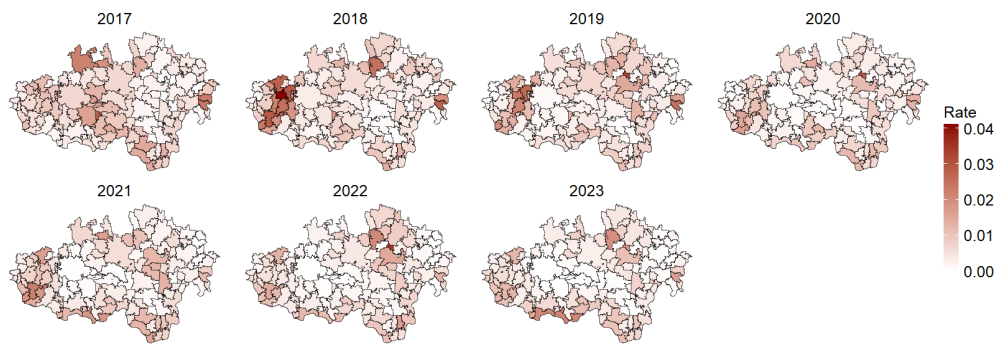


(b)

Figure 12: Simulated ZCTA-level rates of naloxone administrations, zoomed to show the ZCTA that intersect central Ohio counties, for the 8th (Figure a) and 23rd (Figure b) simulated data sets.



(a)



(b)

Figure 13: Simulated ZCTA-level rates of naloxone administrations, zoomed to show the ZCTA that intersect southern Ohio counties, for the 8th (Figure a) and 23rd (Figure b) simulated data sets.

3.3 Maps of Estimated Atom-Level Rates of Misuse

This section shows the corresponding maps of the estimated atom-level prevalence of misuse, $\hat{N}_t(A_{ij})/P_t(A_{ij})$, resulting from the proposed MIA model. The results shown here are the result of fitting the proposed model to the 8th and 23rd simulated data sets shown in the previous section, and the corresponding truth are the maps in Section 3.1.

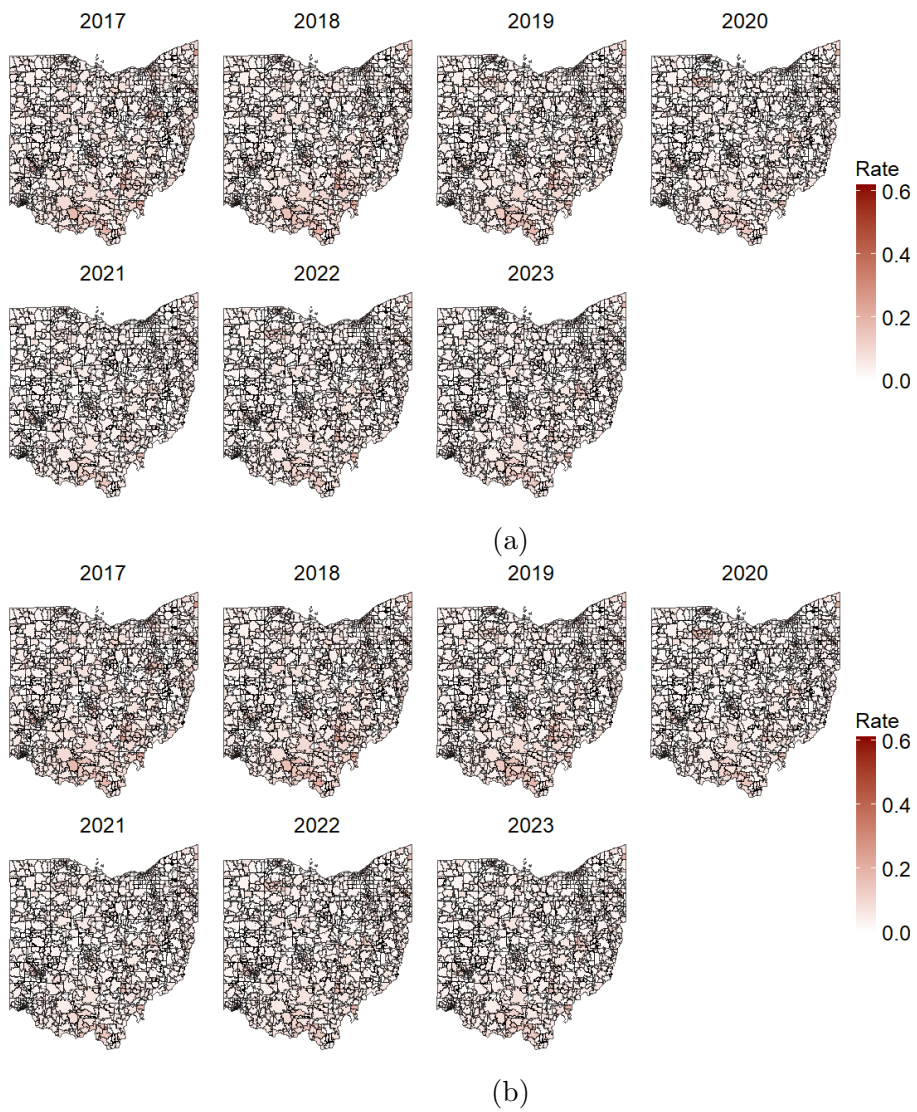
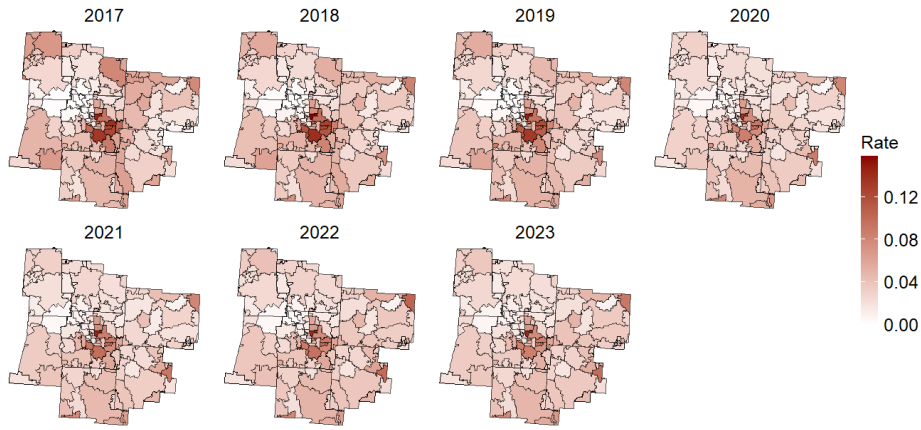
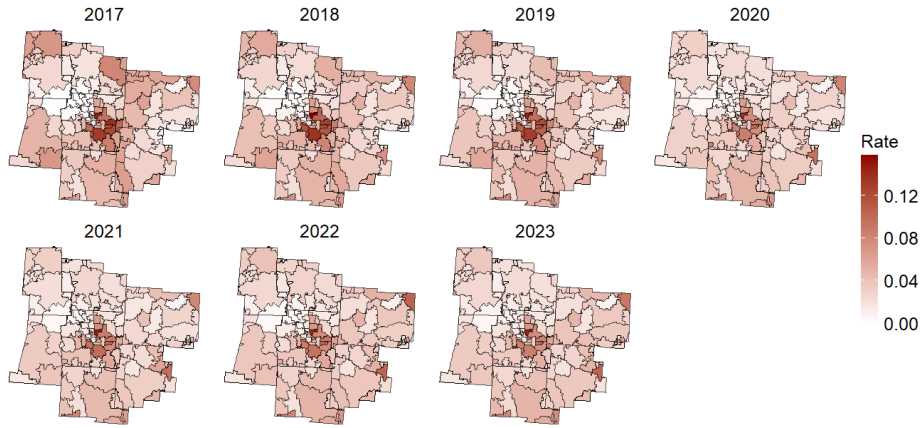


Figure 14: Estimated atom-level prevalence of misuse after fitting the 8th (Figure a) and 23rd (Figure b) simulated data sets.

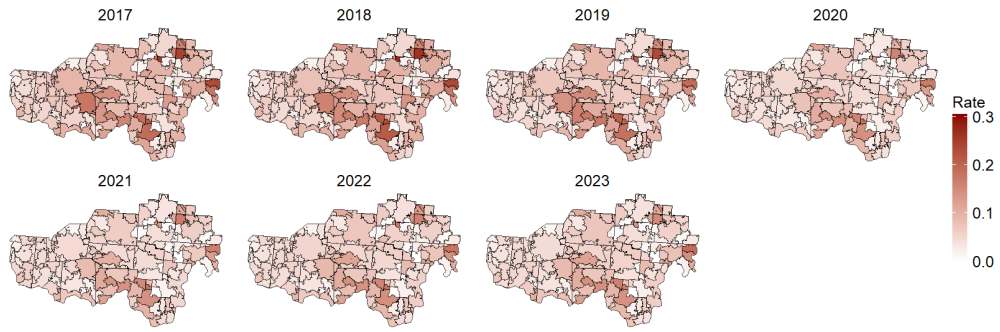


(a)

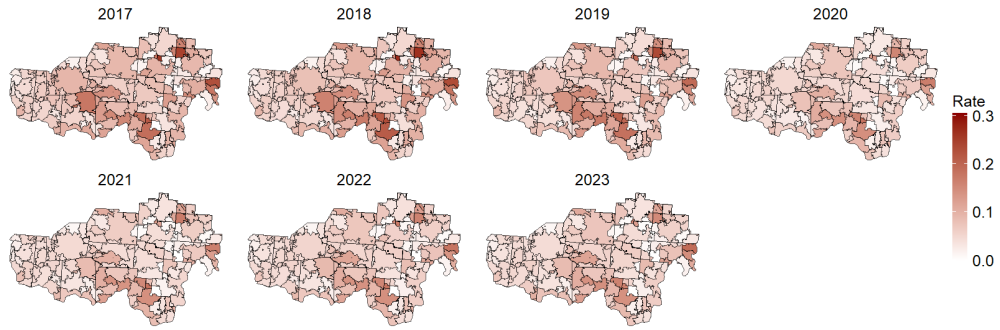


(b)

Figure 15: Estimated atom-level prevalence of misuse, zoomed into the atoms contained within central Ohio counties, after fitting the 8th (Figure a) and 23rd (Figure b) simulated data sets.



(a)



(b)

Figure 16: Estimated atom-level prevalence of misuse, zoomed into the atoms contained within southern Ohio counties, after fitting the 8th (Figure a) and 23rd (Figure b) simulated data sets.

4 Additional Simulation Results

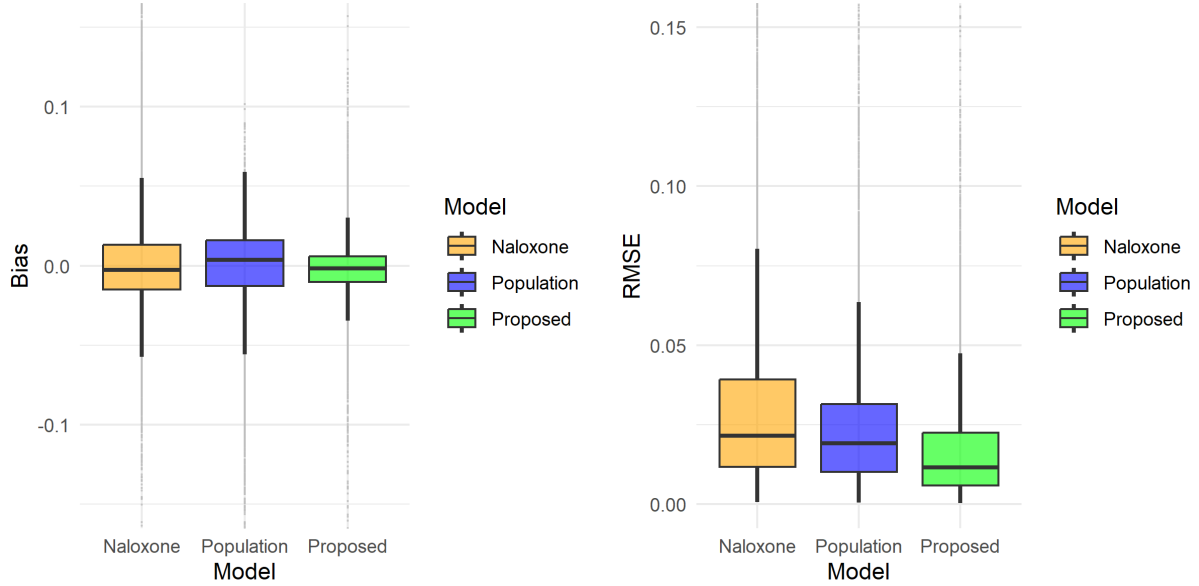


Figure 17: Boxplots of the empirical bias (left) and RMSE (right) for each location-time combination for the proposed MIA model estimates, the comparison estimates that down-scale proportionally to naloxone counts, and the comparison estimates that downscale proportionally to population size.

Figure 17 contains boxplots of the empirical bias and root mean-squared error (RMSE) of the estimated OMP for our proposed MIA model and the two comparison models. The empirical bias is averaged over the 100 simulated data sets and computed as computed as $\text{bias}_{ijt} = \frac{1}{100} \sum_{m=1}^{100} \left(\hat{N}_t(A_{ij})/P_t(A_{ij}) - N_t(A_{ij})/P_t(A_{ij}) \right)$, and the RMSE is given by $\text{RMSE}_{ijt} = \sqrt{\frac{1}{100} \sum_{m=1}^{100} \left(\hat{N}_t(A_{ij})/P_t(A_{ij}) - N_t(A_{ij})/P_t(A_{ij}) \right)^2}$. For the proposed model, the estimated atom-level counts are computed as the median of the posterior distribution, and the boxplots summarize these quantities across all locations and times. Note that outliers are portrayed as light gray dots, and the figure has been zoomed in such that not all outliers appear.

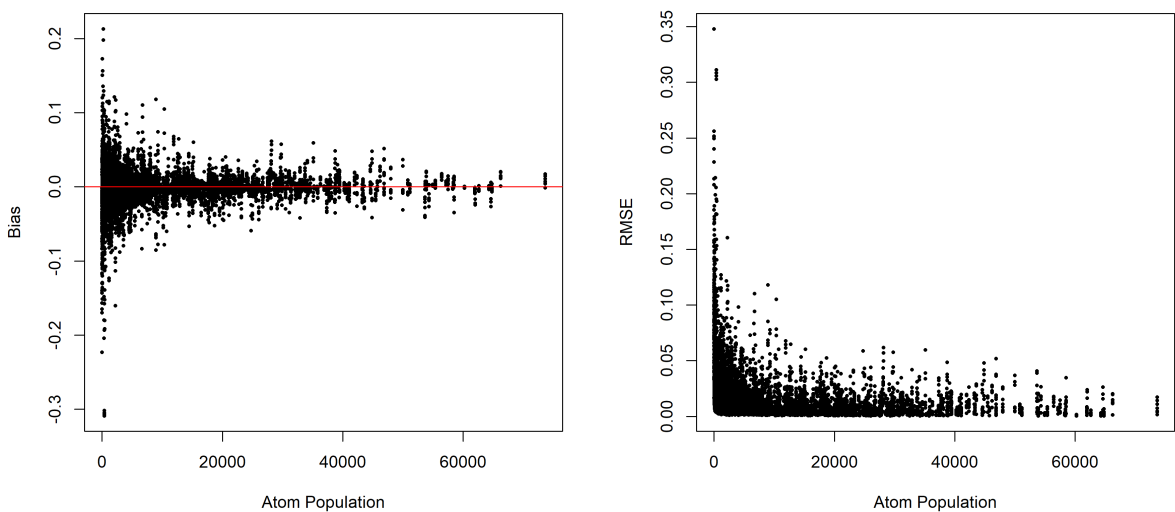


Figure 18: Plot of the empirical bias and RMSE versus the atom-level population. In general, we see the error decreases as the population increases.

5 Supplemental Application Results

5.1 County-Level Model Results

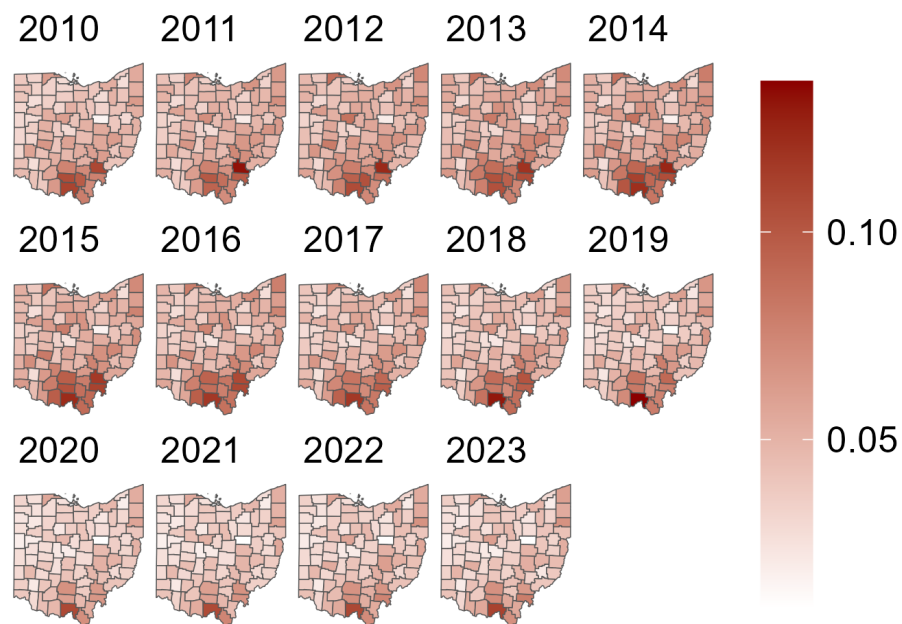


Figure 19: County-level estimated OMP for 2010 to 2023.

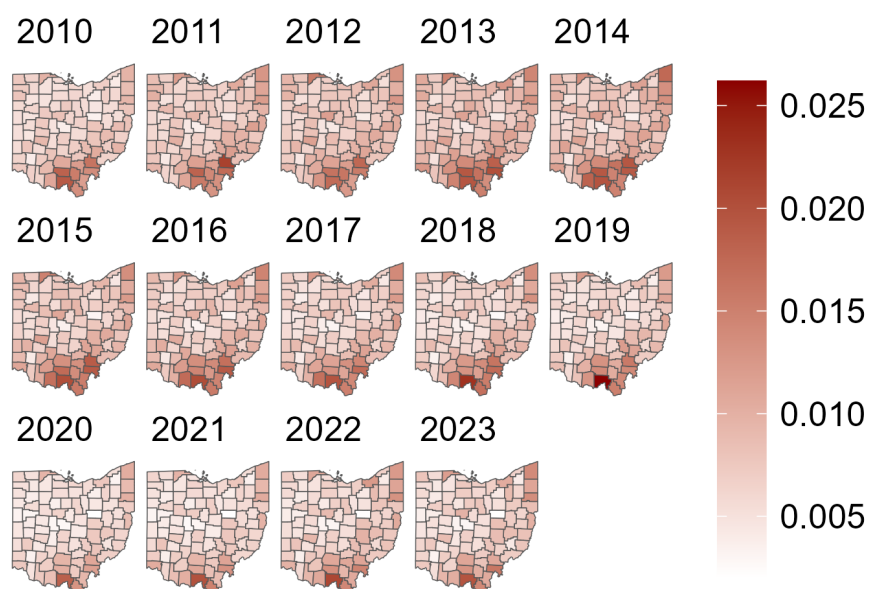


Figure 20: Standard deviation of county-level OMP for 2010 to 2023.

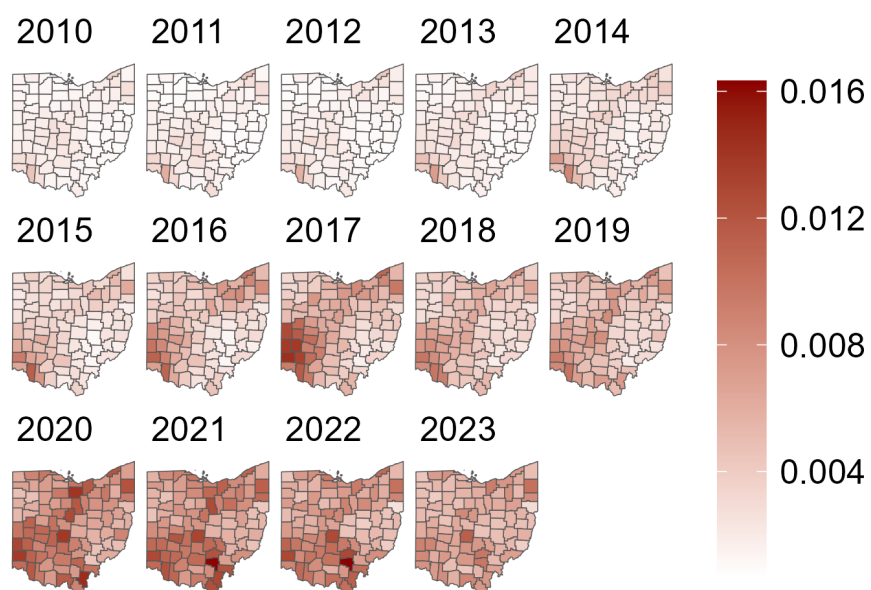


Figure 21: Estimated county-level rate of opioid overdose death among PWMO.

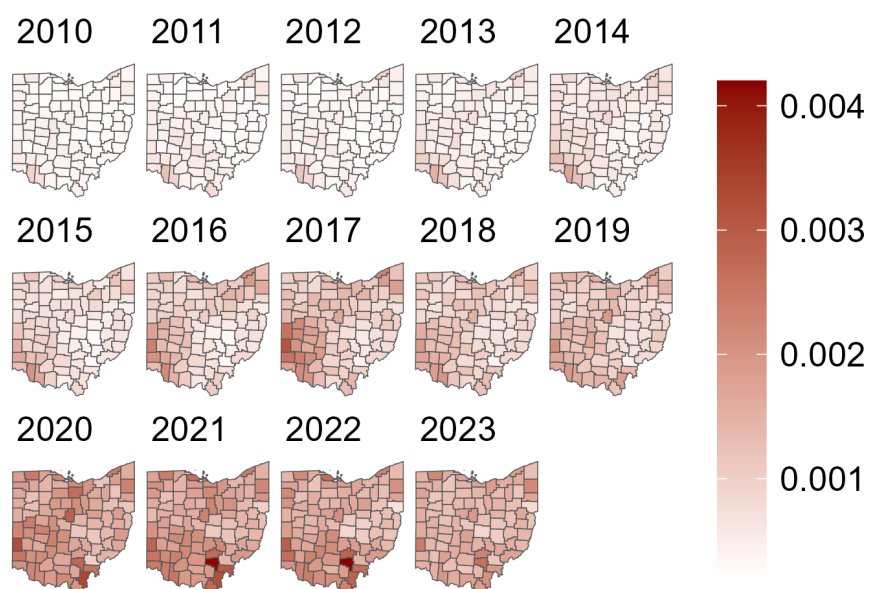


Figure 22: Standard deviation of the rate of opioid overdose death among PWMO.

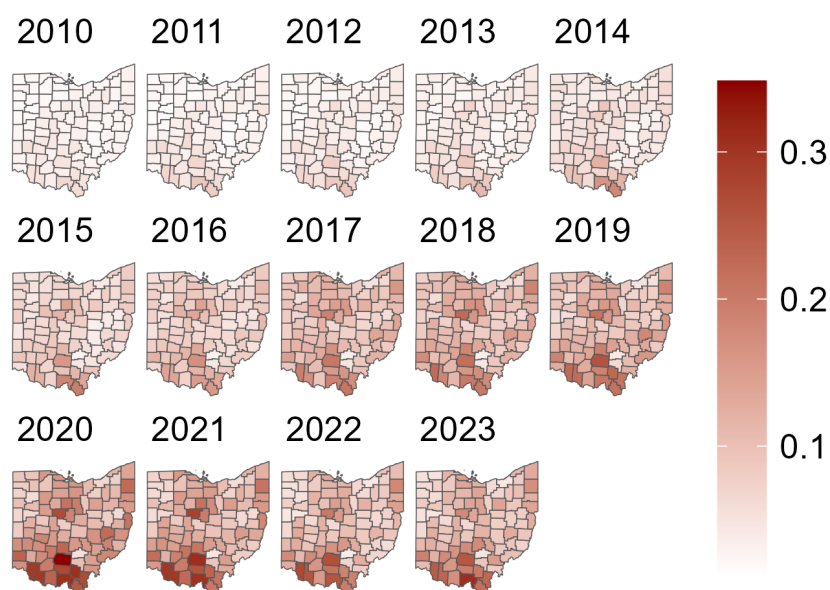


Figure 23: Estimated county-level rate of opioid treatment admissions among PWMO.

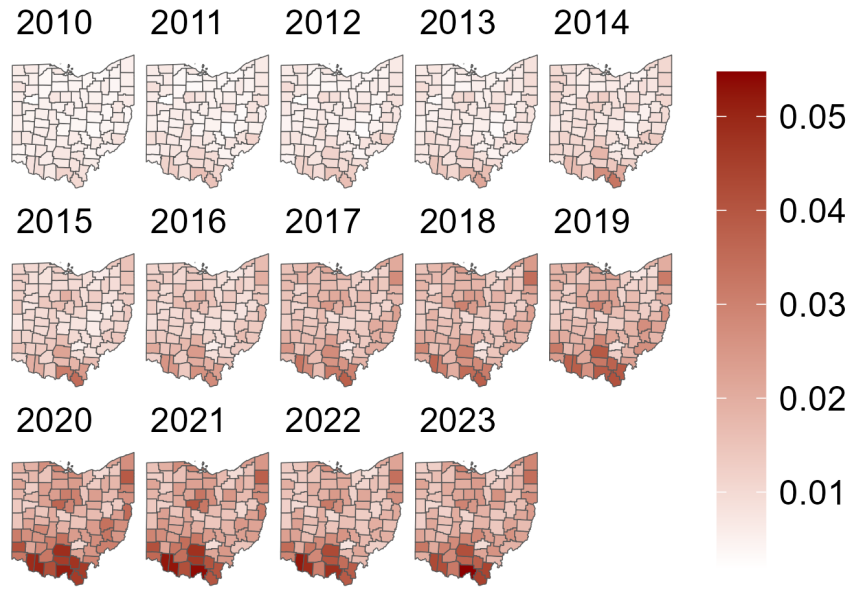


Figure 24: Standard deviation of the rate of opioid treatment admissions among PWMO.

	Parameter	Mean	95% CI
Treatment Rate	HPSA	-0.02	(-0.08, 0.04)
	MU	0.04	(-0.03, 0.1)
Death Rate	Interstates	0.05	(-0.03, 0.13)
	HIDTA	0.02	(-0.06, 0.11)
	MSA	0.19	(0.1, 0.27)
Misuse Prevalence	Poverty	0.18	(0.13, 0.22)
	Unemployed	0.02	(-0.02, 0.07)
	High School Degree	0.13	(0.11, 0.17)
	Food Stamps	0.31	(0.27, 0.36)
	Prescribing Rate	-0.05	(-0.08, -0.02)

Table 2: Estimates and 95% credible intervals for the regression coefficients in the outcome-level models for the rates of treatment and death among those who misuse opioids, as well as the covariate effects of the county-level prevalence of misuse.

5.2 Sub-County Level Model Results

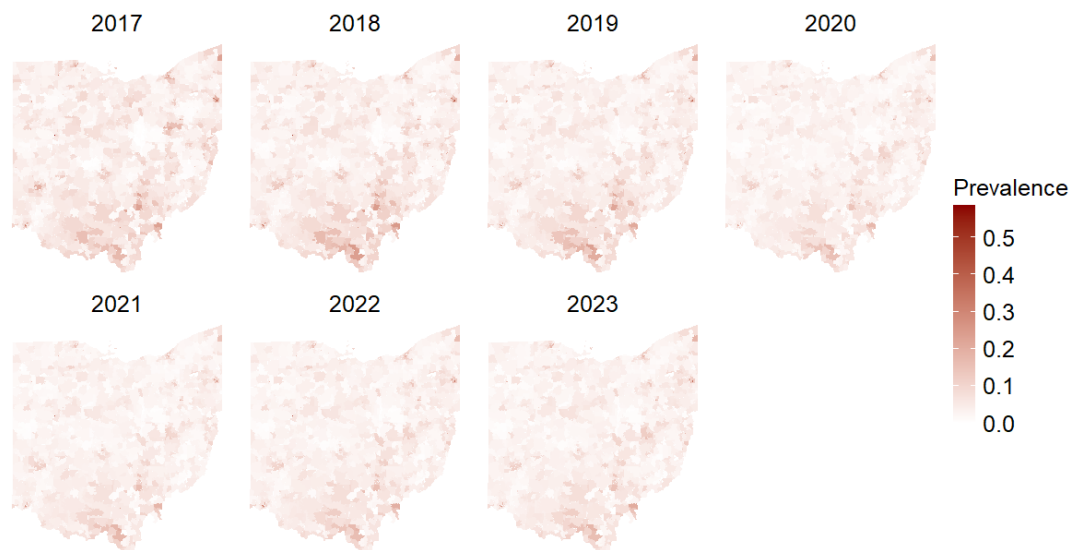


Figure 25: Estimated atom-level OMP.

Parameter	Mean	CI
Intercept	-3.08	(-3.14, -3.03)
Metropolitan	0.57	(0.5, 0.63)
Micropolitan	0.01	(-0.03, 0.05)

Table 3: Estimates and 95% credible intervals for the regression coefficients in the ZCTA-level model of the rate of naloxone administrations among people who misuse opioids.

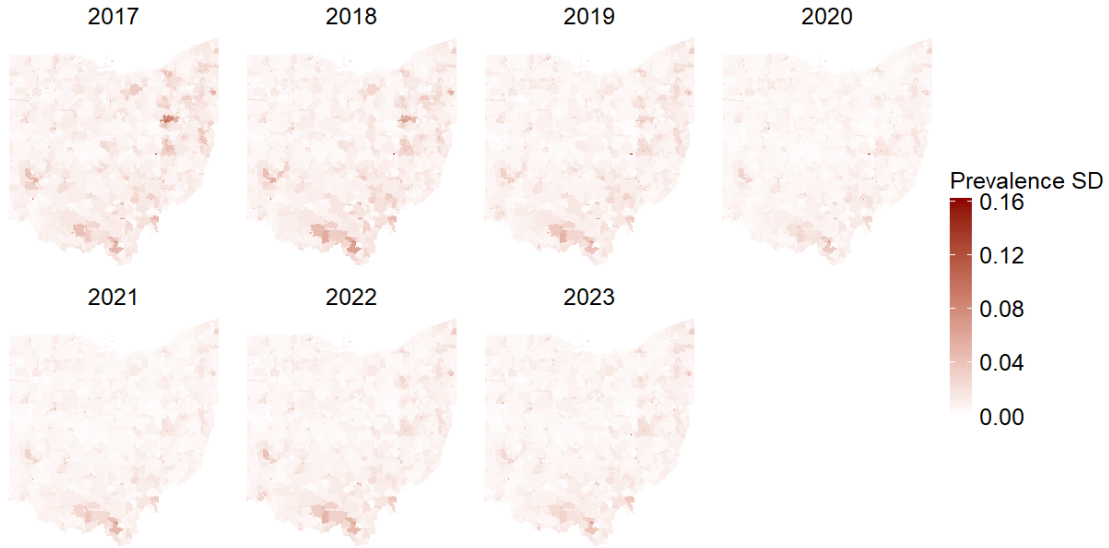


Figure 26: Posterior standard deviation of atom-level OMP.

Parameter	Mean	95% CI
Intercept	-0.77	(-1.01, -0.49)
Unemployment	0.98	(0.9, 1.06)
College Degree	-0.28	(-0.34, -0.22)

Table 4: Estimates and 95% credible intervals for the regression coefficients in the atom-level model of prevalence of people who misuse opioids.

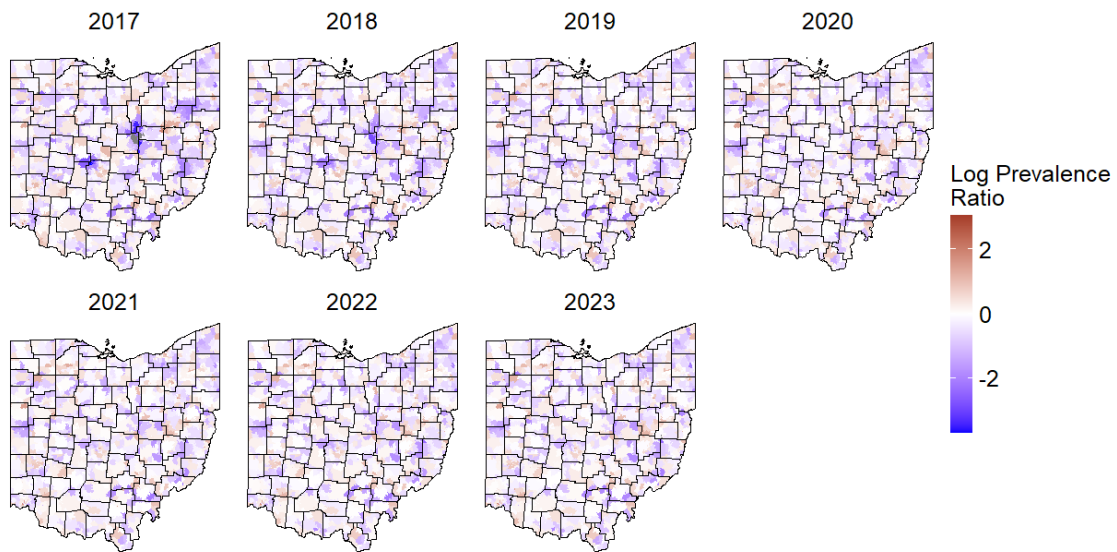


Figure 27: Log prevalence ratio of atom-level OMP relative to the county-level OMP.

5.3 State-Level Results

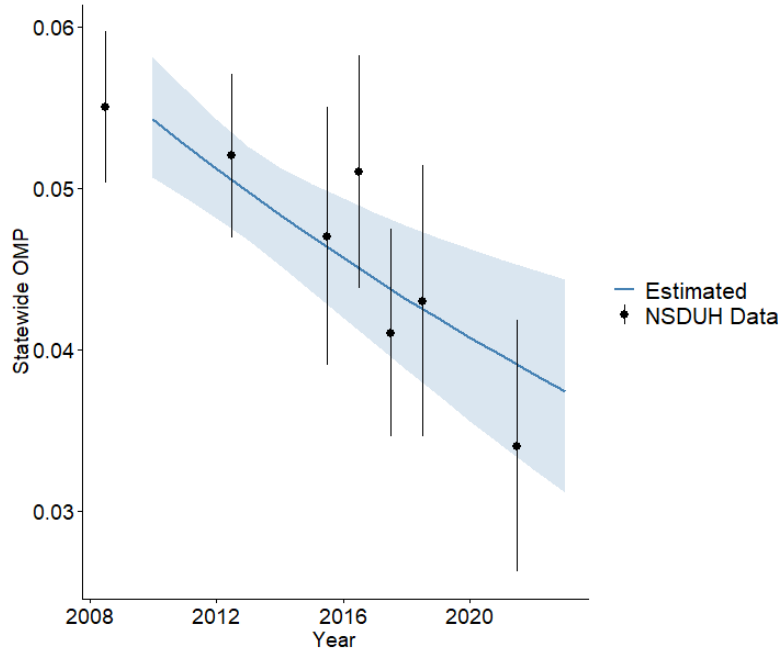


Figure 28: Estimated state-level OMP with the 95% credible interval shown as the shaded region. The NSDUH survey estimates are shown as points plotted at the midpoint of the time period of the survey. The error bars are 1.95 standard deviations above and below the estimate, using the standard deviations provided by NSDUH.

References

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